Block practice · Inheritance and genetic diagrams

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Define a gene mutation and name its causes

Explain how a mutation changes a protein and phenotype

Draw a genetic diagram for a monohybrid or sex-linked cross

Explain sex-linked inheritance and why males are affected more

Deduce genotypes from a pedigree, family or blood group

Build or read a pedigree to decide how a condition is inherited

Work out the probability of a child's genotype or phenotype from a cross

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Practice workbook · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Defining and using gene, allele, dominant, recessive, homozygous, heterozygous, genotype, phenotype and codominance; listing the genotypes for a blood group; and explaining how carriers pass on a recessive condition.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-GENETIC-TERMS

1. Question 7 · part a.i · 2 marks

June 2023 Paper 1R

Worked-example retry · Attempt part a.i only.

7 The four blood groups that occur in humans are group A, group B, group AB and group O.

There are three blood group alleles: I^A, I^B and I^O.

I^O is recessive to both I^A and I^B.

I^A and I^B are co-dominant to each other.

(a) (i) Explain what is meant by the term **co-dominant**.

(2)

(ii) State the possible genotypes of a person with blood group B.

(2)

(b) Haemophilia is a sex-linked condition that prevents the blood clotting.

Explain why males are more likely to be affected by haemophilia than females.

(4)

(Total for Question 7 = 8 marks)

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1. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 9 · part a.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.i only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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P 5 8 5 6 8 A 0 2 4 2 8

2. Question 9 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part a.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part a.i only.

8 (a) Genetic conditions are often caused by recessive alleles.

(i) Which of these describes an allele?

(1)

- ☒ **A** a chromosome that controls a characteristic
- ☒ **B** an alternative form of a gene
- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

.....

.....

(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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P 6 7 0 6 2 R A 0 2 6 2 8

3. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 8 · part a.ii · 1 mark

November 2021 Paper 2

Further practice · Attempt part a.ii only.

8 (a) Genetic conditions are often caused by recessive alleles.

(i) Which of these describes an allele?

(1)

- ☒ **A** a chromosome that controls a characteristic
- ☒ **B** an alternative form of a gene
- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

.....

.....

(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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P 6 7 0 6 2 R A 0 2 6 2 8

4. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 5 · part a · 2 marks

November 2023 Paper 2

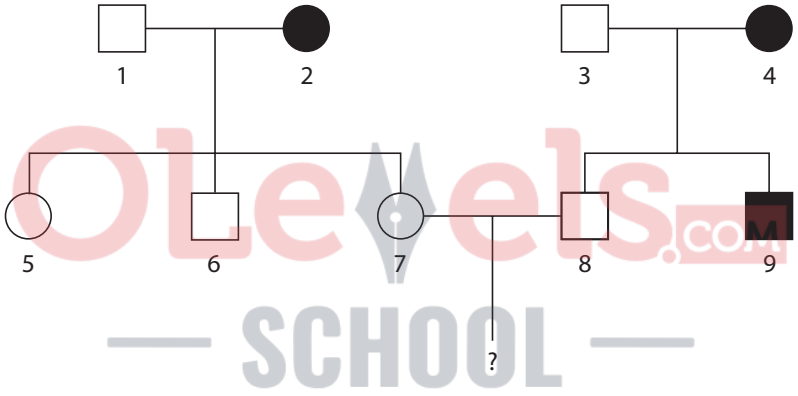
Further practice · Attempt part a only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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5. Question 5 · part a · 2 marks · Working and answer

Working and answer

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6. Question 3 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

.....

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

.....

(iii) Give one symptom of polydactyly.

(1)

.....

.....

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

.....

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P 7 4 6 2 6 R A 0 8 2 4

6. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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7. Question 3 · part b.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.i only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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7. Question 3 · part b.i · 1 mark

June 2024 Paper 1

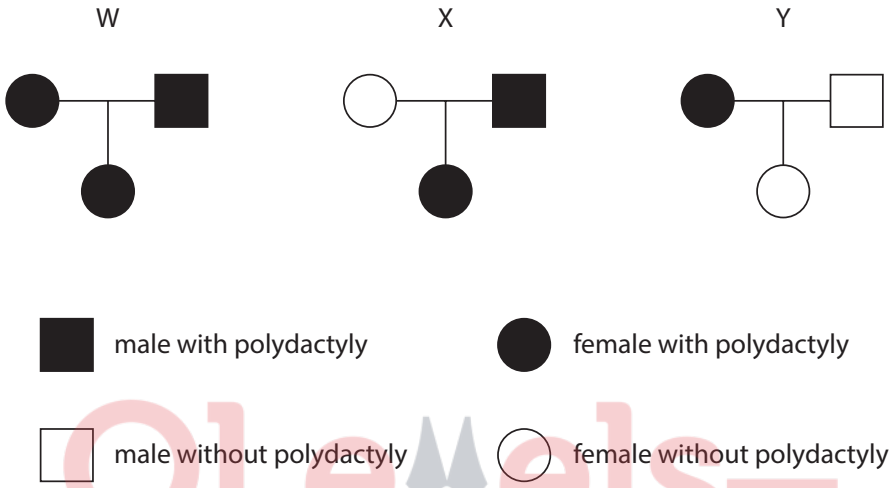
Further practice · Attempt part b.i only.

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(ii) The family pedigree diagrams show the inheritance of polydactyly.



Which of the diagrams, W, X or Y, gives the family pedigree for parents R and S?

(1)

(c) A gene is a short section of DNA.

Describe the structure of DNA.

(4)

(Total for Question 3 = 9 marks)

7. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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8. Question 4 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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P 7 4 4 9 5 A 0 1 2 2 8

8. Question 4 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

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(ii) Explain why diploid cells contain two alleles for each characteristic.

(4)

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

(2)

(ii) Give one symptom of cystic fibrosis.

(1)

(Total for Question 4 = 11 marks)

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8. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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9. Question 6 · part a · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a only.

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6 (a) Individuals vary in their phenotypes.

State what is meant by the phenotype of an individual.

(1)

(b) (i) Cystic fibrosis is a disorder that affects epithelial membranes.

Describe what causes an individual to have cystic fibrosis.

(2)

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(ii) Describe the effects of cystic fibrosis on the health of an individual.

(4)

9. Question 6 · part a · 1 mark · Working and answer

Working and answer

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10. Question 9 · part a.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.ii only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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10. Question 9 · part a.ii · 1 mark · Working and answer

Working and answer

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11. Question 7 · part a.ii · 2 marks

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

7 The four blood groups that occur in humans are group A, group B, group AB and group O.

There are three blood group alleles: I^A, I^B and I^O.

I^O is recessive to both I^A and I^B.

I^A and I^B are co-dominant to each other.

(a) (i) Explain what is meant by the term **co-dominant**.

(2)

(ii) State the possible genotypes of a person with blood group B.

(2)

(b) Haemophilia is a sex-linked condition that prevents the blood clotting.

Explain why males are more likely to be affected by haemophilia than females.

(4)

(Total for Question 7 = 8 marks)

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11. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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12. Question 8 · part a.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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P 6 2 0 5 2 A 0 1 8 2 4

12. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

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13. Question 8 · part b.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

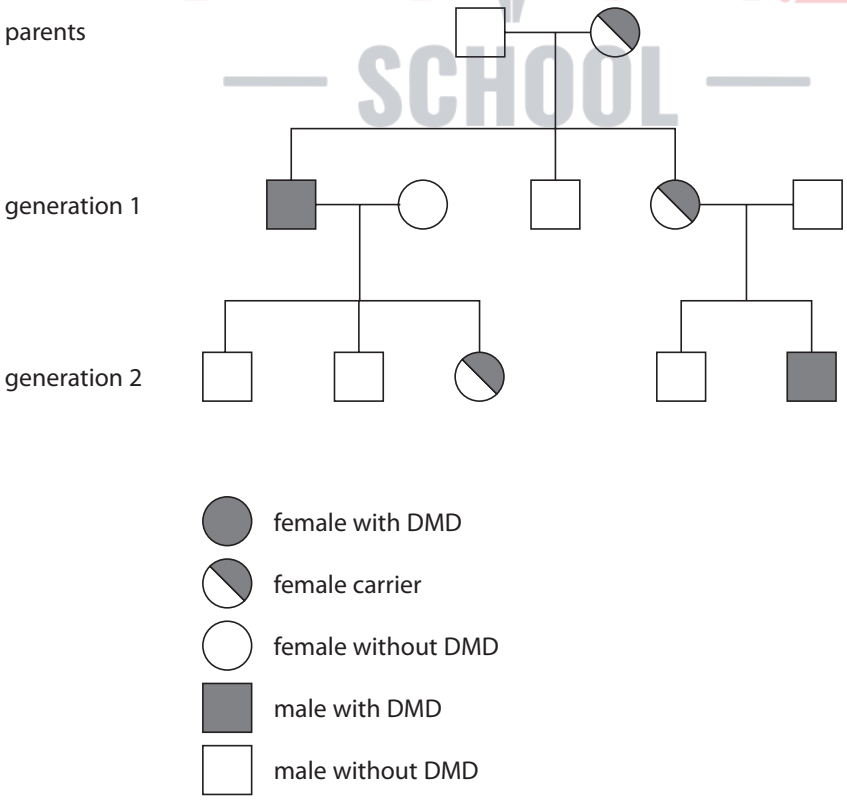
8 (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

(4)

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.



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13. Question 8 · part b.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

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(i) Explain what is meant by a carrier of DMD.

(2)

(ii) Explain how the diagram shows that DMD is caused by a recessive allele.

(3)

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(iii) Draw a genetic diagram to show the inheritance of DMD from the parents to generation 1.

Use X^D for the dominant allele and X^d for the recessive allele.

(2)

(Total for Question 8 = 11 marks)

13. Question 8 · part b.i · 2 marks · Working and answer

Working and answer

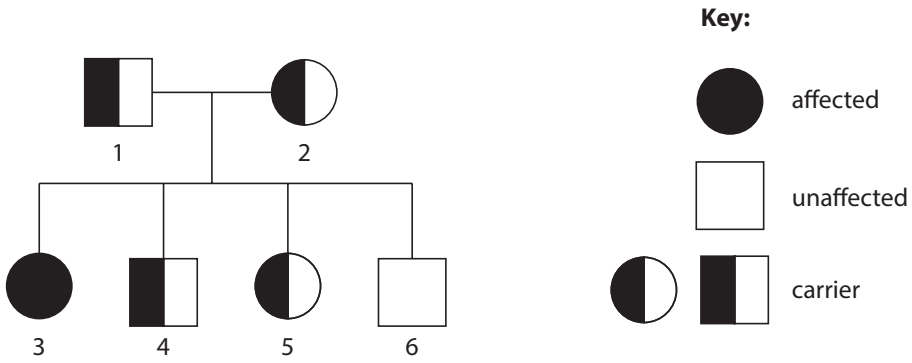
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14. Question 8 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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14. Question 8 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

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(iv) Draw a genetic diagram to show how two parents can have a 1 in 4 chance of having a child with cystic fibrosis.

Use F for the non-cystic fibrosis allele and f for the cystic fibrosis allele.

(4)

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(b) Describe how gene therapy can be used to treat cystic fibrosis.

(3)

(Total for Question 8 = 11 marks)

14. Question 8 · part a.iii · 2 marks · Working and answer

Working and answer

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15. Question 6 · part b.i · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

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6 (a) Individuals vary in their phenotypes.

State what is meant by the phenotype of an individual.

(1)

(b) (i) Cystic fibrosis is a disorder that affects epithelial membranes.

Describe what causes an individual to have cystic fibrosis.

(2)

(ii) Describe the effects of cystic fibrosis on the health of an individual.

(4)

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15. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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16. Question 6 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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6 (a) Individuals vary in their phenotypes.

State what is meant by the phenotype of an individual.

(1)

(b) (i) Cystic fibrosis is a disorder that affects epithelial membranes.

Describe what causes an individual to have cystic fibrosis.

(2)

(ii) Describe the effects of cystic fibrosis on the health of an individual.

(4)

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16. Question 6 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(iii) Explain why two parents who do not have cystic fibrosis can have a child who has cystic fibrosis.

(2)

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(Total for Question 6 = 9 marks)

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Working and answer

Practice workbook · Define a gene mutation and name its causes

Stating what a mutation is, how it alters a chromosome, and naming mutagens or why a tissue such as skin is prone.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-MUTATION-AND-CAUSES

1. Question 5 · part a.i · 2 marks

November 2024 Paper 2

Worked-example retry · Attempt part a.i only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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1. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

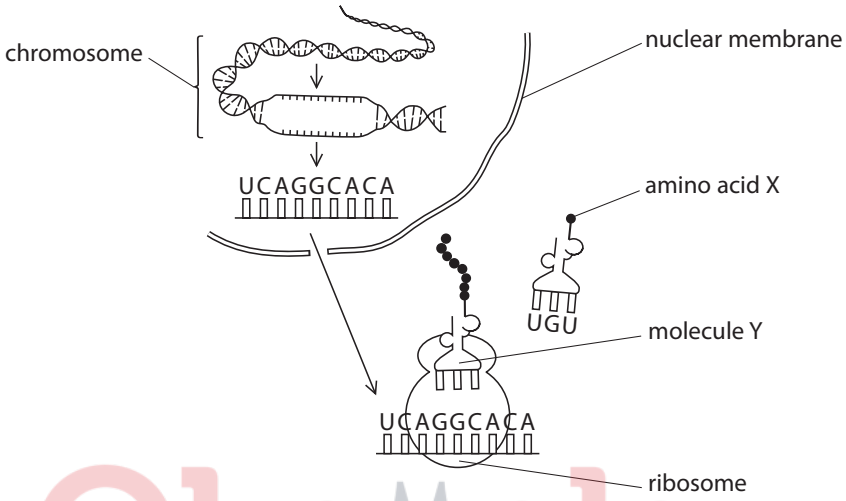
2. Question 6 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

6 Protein synthesis has a number of stages.

The diagram shows some of the stages.



The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

Base triplet of DNA	Amino acid
AGT	serine
CCG	glycine
TGT	threonine
GTA	histidine
CAA	valine
TCC	argenine
ACA	cysteine

(a) Describe the function of the nuclear membrane.

(2)

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2. Question 6 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

(1)

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

(1)

(d) Describe the process occurring on the ribosome.

(6)

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2. Question 6 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

(e) Describe how a mutation will affect a chromosome.

(2)

(Total for Question 6 = 12 marks)

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Working and answer

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3. Question 8 · part a · 3 marks

November 2020 Paper 2

Further practice · Attempt part a only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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3. Question 8 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 3 · part a · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a only.

3 (a) Polycystic kidney disease (PKD) is an inherited condition caused by a gene mutation.

PKD affects the function of the kidneys.

State what is meant by a gene mutation.

(1)

.....

.....

(b) PKD occurs when a child inherits a faulty allele from one parent.

One parent has a genotype Hh and the other parent has a genotype hh.

The parents have four children.

Draw a genetic diagram to show the possible genotypes of the four children.

(2)

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(c) Some people with PKD need a kidney transplant.

Give one advantage and one disadvantage of a kidney transplant.

(2)

Advantage

.....

.....

Disadvantage

.....

.....

(Total for Question 3 = 5 marks)

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4. Question 3 · part a · 1 mark · Working and answer

Working and answer

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5. Question 5 · part b.ii · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.ii only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

5. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

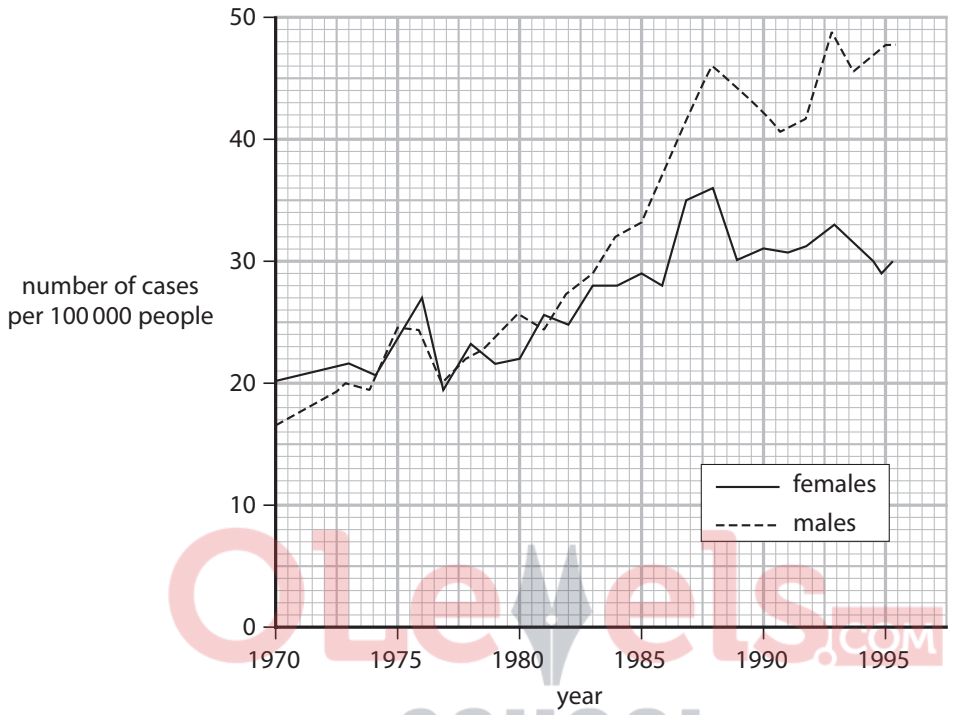
OLevels.com SCHOOL

6. Question 6 · part a.ii · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

6 A form of skin cancer called melanoma has been increasing in both males and females.



(a) (i) Describe the trends shown by the graph.

(3)

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P 6 7 0 6 0 R R A 0 1 4 2 0

6. Question 6 · part a.ii · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Suggest why cancer can occur in the skin.

(2)

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

(3)

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mean increase per 100 000 males per year = ...

(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

(3)

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number of females with melanoma = ...

(Total for Question 6 = 11 marks)

6. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how a mutation changes a protein and phenotype

Chaining a base change through codons and amino acids to a faulty or missing protein and a named condition or characteristic.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-MUTATION-TO-PHENOTYPE

1. Question 7 · part a · 3 marks

January 2023 Paper 1

Worked-example retry · Attempt part a only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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P 7 2 2 7 1 A 0 2 0 2 4

1. Question 7 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part c · 4 marks

June 2021 Paper 2

Further practice · Attempt part c only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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P 6 6 4 3 3 A 0 2 0 2 4

2. Question 8 · part c · 4 marks

June 2021 Paper 2

Further practice · Attempt part c only.

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(ii) Calculate the probability that these parents will produce a boy with the condition.

(3)

probability = ...

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(c) White forelock is a result of a mutation in a molecule of DNA.

Describe how this mutation causes the white forelock phenotype.

(4)

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(Total for Question 8 = 14 marks)

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2. Question 8 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 5 · part a · 3 marks

January 2022 Paper 1

Further practice · Attempt part a only.

5 (a) Describe the possible effects of a DNA mutation.

(3)

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

Age of mother in years	15–19	20–24	25–29	30–34	35–39	40–44
Probability of chromosome aberration	0.0010	0.0010	0.0010	0.0015	0.0045	0.0140

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3. Question 5 · part a · 3 marks · Working and answer

Working and answer

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4. Question 8 · part a · 4 marks

June 2022 Paper 2

Further practice · Attempt part a only.

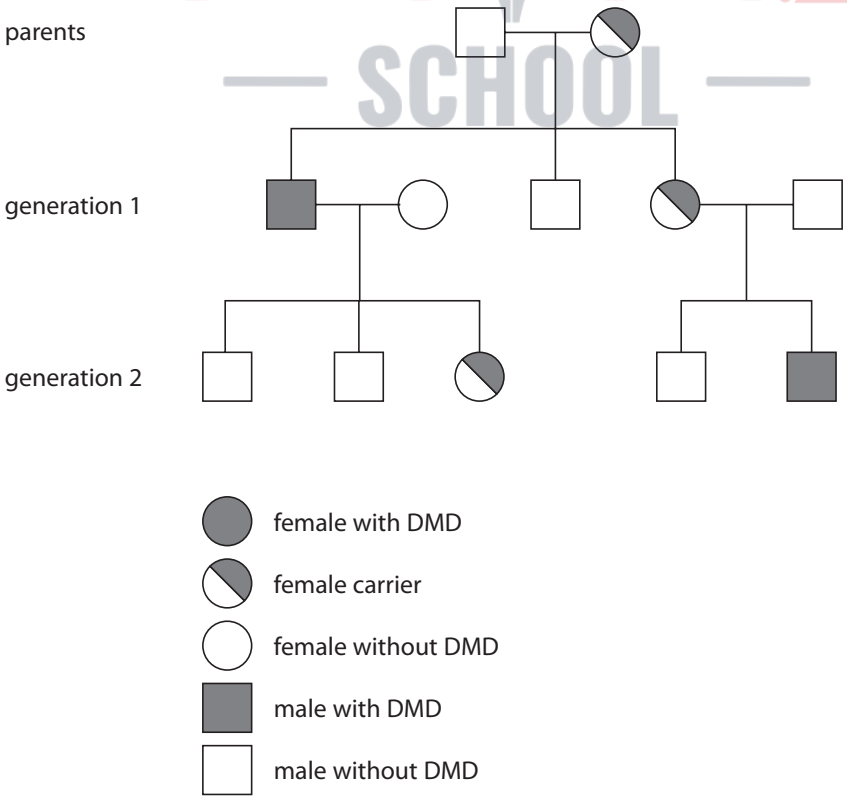
8 (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

(4)

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.



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4. Question 8 · part a · 4 marks · Working and answer

Working and answer

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5. Question 6 · part a.i · 4 marks

June 2023 Paper 2

Further practice · Attempt part a.i only.

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6 Sickle cell anaemia is an inherited condition. It is caused by a mutation of the gene that controls the production of haemoglobin.

This condition is caused by a recessive allele.

(a) (i) Explain how a mutation can cause a condition such as sickle cell anaemia.

(4)

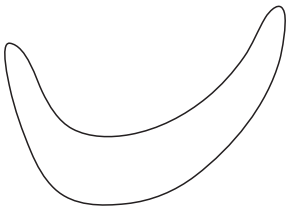
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5. Question 6 · part a.i · 4 marks

June 2023 Paper 2

Further practice · Attempt part a.i only.

(ii) In low oxygen concentrations the red blood cells of someone with sickle cell anaemia become like the cell shown in the diagram.



Explain why a person who has the sickle cell condition often feels tired.

(5)

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5. Question 6 · part a.i · 4 marks · Working and answer

Working and answer

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6. Question 5 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

6. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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7. Question 5 · part b.iv · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.iv only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

7. Question 5 · part b.iv · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.iv only.

(iv) State why some mutations have no effect on the phenotype of an organism.

(1)

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(Total for Question 5 = 9 marks)

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Working and answer

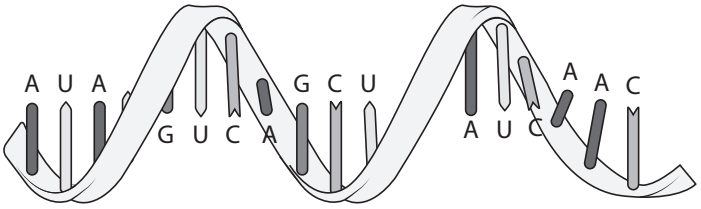
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8. Question 2 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

2 (a) The diagram shows a strand of RNA found in a body cell.



(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

(2)

1 .....

2 .....

(ii) State two differences between RNA and DNA.

(2)

1 .....

.....

2 .....

.....

(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

(1)

.....

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P 7 4 6 3 0 A 0 6 2 4

8. Question 2 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(b) Before a cell divides by mitosis, the DNA inside the cell is replicated.

When DNA is replicated, mutations can happen.

Explain how mutations can affect the final product made.

(3)

(Total for Question 2 = 8 marks)

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Working and answer

9. Question 5 · part a.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part a.ii only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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P 7 5 9 5 9 A 0 1 5 2 4

Turn over ▶

9. Question 5 · part a.ii · 3 marks

November 2024 Paper 2

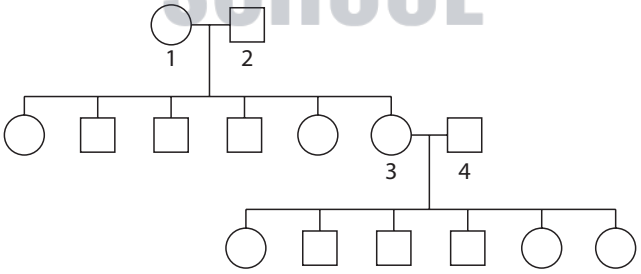
Further practice · Attempt part a.ii only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

(2)

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.



Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

(2)

person 1

person 2

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9. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

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10. Question 7 · part a.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.i only.

7 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

(2)

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

(2)

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10. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Draw a genetic diagram for a monohybrid or sex-linked cross

Full crosses for autosomal, codominant blood group, X-linked conditions and sex determination by X and Y chromosomes.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DRAW-MONOHYBRID-GENETIC-DIAGRAM

1. Question 5 · part c.i · 4 marks

November 2023 Paper 2

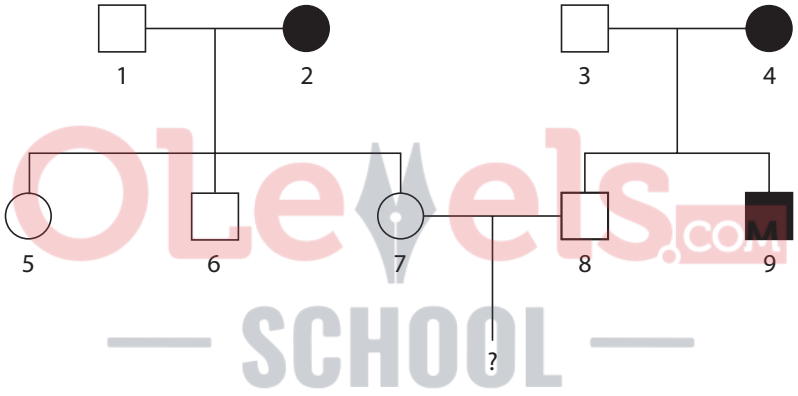
Worked-example retry · Attempt part c.i only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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1. Question 5 · part c.i · 4 marks

November 2023 Paper 2

Worked-example retry · Attempt part c.i only.

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(c) (i) People 7 and 8 in the pedigree produce a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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(ii) Calculate the probability that this child will be a boy with sickle cell anaemia.

(4)

probability = ...

(Total for Question 5 = 13 marks)

1. Question 5 · part c.i · 4 marks · Working and answer

Working and answer

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2. Question 9 · part c.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part c.i only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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P 5 8 5 6 8 A 0 2 4 2 8

2. Question 9 · part c.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part c.i only.

(c) Haemophilia is a sex-linked blood disorder that reduces the ability of the blood to clot.

These are the genotypes of four offspring, P, Q, R and S.

P	Q	R	S
X^HX^h	X^hY	X^HX^H	X^HY

(i) Draw a genetic diagram to show how these offspring are produced from one set of parents.

(2)

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(ii) These parents are expecting another baby.

Determine the probability that this baby will have haemophilia.

(1)

probability = ...

(Total for Question 9 = 9 marks)

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2. Question 9 · part c.i · 2 marks · Working and answer

Working and answer

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3. Question 8 · part b.i · 4 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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3. Question 8 · part b.i · 4 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

(ii) The man and woman have a second child.

Calculate the probability that both children will have haemophilia.

(3)

(c) Explain why the man cannot be a carrier of haemophilia.

(5)

(Total for Question 8 = 15 marks)

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3. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 8 · part b.i · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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4. Question 8 · part b.i · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

(ii) Calculate the probability that these parents will produce a boy with the condition.

(3)

probability = ...

(c) White forelock is a result of a mutation in a molecule of DNA.

Describe how this mutation causes the white forelock phenotype.

(4)

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(Total for Question 8 = 14 marks)

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4. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 7 · part b.ii · 6 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

7 Huntington's chorea is a dominant genetic condition that results in the nervous system not working correctly. It is not a sex-linked condition.

Diagram 1 shows a family pedigree. Some members of the family have the condition.

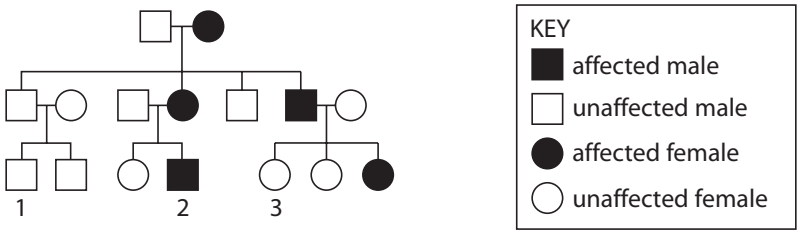


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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5. Question 7 · part b.ii · 6 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

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(b) (i) A man has the same genotype as person 1 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

These two people are expecting a child.

Calculate the probability that the child would be a boy not showing the Huntington's condition, giving reasons for your answer.

(3)

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5. Question 7 · part b.ii · 6 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

(ii) A man has the same genotype as person 2 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

Explain the possible phenotypes and genotypes, with respect to the Huntington's condition, of any offspring they produce.

(6)

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(Total for Question 7 = 13 marks)

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5. Question 7 · part b.ii · 6 marks · Working and answer

Working and answer

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6. Question 4 · part a.iii · 3 marks

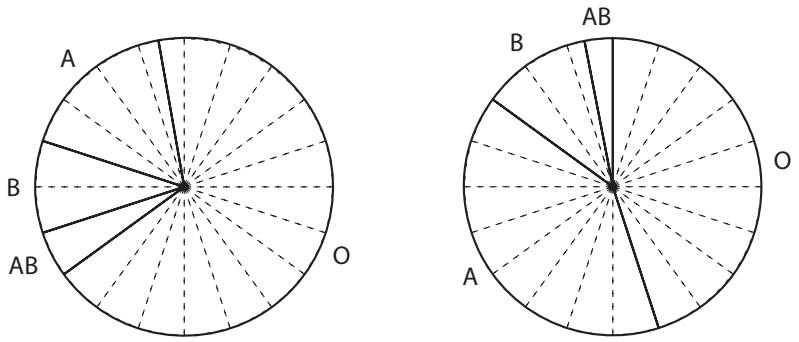
June 2022 Paper 1

Further practice · Attempt part a.iii only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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P 7 0 9 7 3 A 0 1 2 2 4

6. Question 4 · part a.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...

6. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b.iii · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.iii only.

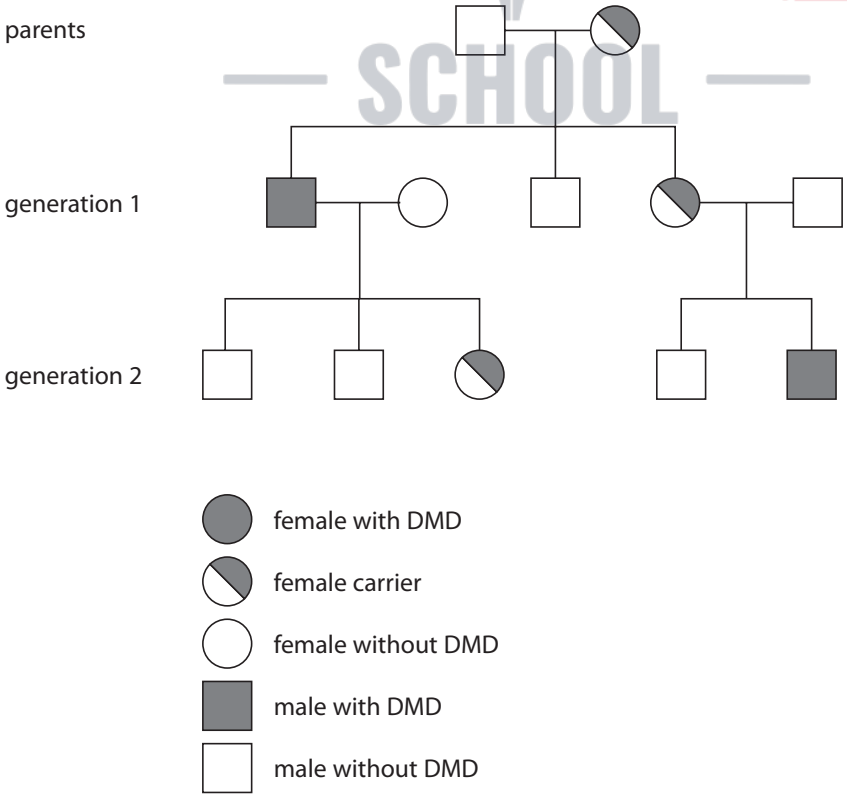
8 (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

(4)

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.



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7. Question 8 · part b.iii · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.iii only.

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(i) Explain what is meant by a carrier of DMD.

(2)

(ii) Explain how the diagram shows that DMD is caused by a recessive allele.

(3)

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(iii) Draw a genetic diagram to show the inheritance of DMD from the parents to generation 1.

Use X^D for the dominant allele and X^d for the recessive allele.

(2)

(Total for Question 8 = 11 marks)

7. Question 8 · part b.iii · 2 marks · Working and answer

Working and answer

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8. Question 7 · part c.ii · 4 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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8. Question 7 · part c.ii · 4 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

(2)

percentage = ... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

(2)

1 .....

.....

2 .....

.....

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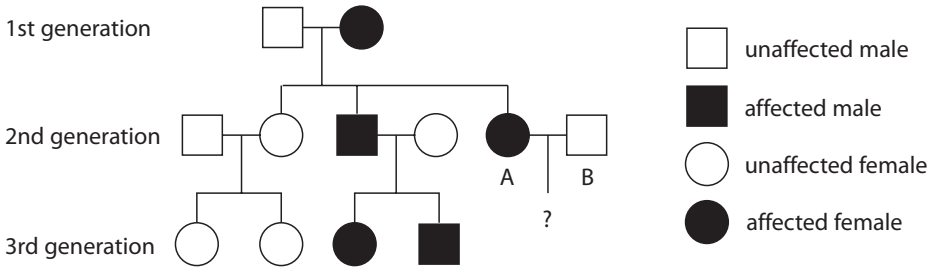
8. Question 7 · part c.ii · 4 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(c) The diagram shows a family pedigree. Some of the family have polycystic kidney disease.

This disease is caused by a dominant allele that is not sex-linked.



(i) Determine the ratio of the females shown in the 3rd generation that are affected with polycystic kidney disease.

(1)

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ratio = ...

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8. Question 7 · part c.ii · 4 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

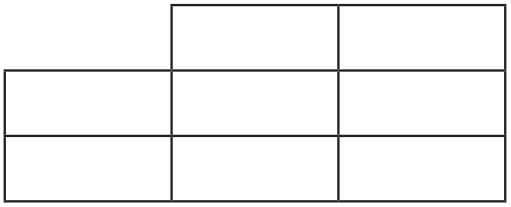
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(ii) Explain the possible genotypes and phenotypes for the offspring of individual A and individual B.

You should complete the genetic diagram as part of your answer.

(4)



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(Total for Question 7 = 14 marks)

8. Question 7 · part c.ii · 4 marks · Working and answer

Working and answer

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9. Question 3 · part b · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

3 (a) Polycystic kidney disease (PKD) is an inherited condition caused by a gene mutation.

PKD affects the function of the kidneys.

State what is meant by a gene mutation.

(1)

.....

.....

(b) PKD occurs when a child inherits a faulty allele from one parent.

One parent has a genotype Hh and the other parent has a genotype hh.

The parents have four children.

Draw a genetic diagram to show the possible genotypes of the four children.

(2)

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(c) Some people with PKD need a kidney transplant.

Give one advantage and one disadvantage of a kidney transplant.

(2)

Advantage

.....

.....

Disadvantage

.....

.....

(Total for Question 3 = 5 marks)

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9. Question 3 · part b · 2 marks · Working and answer

Working and answer

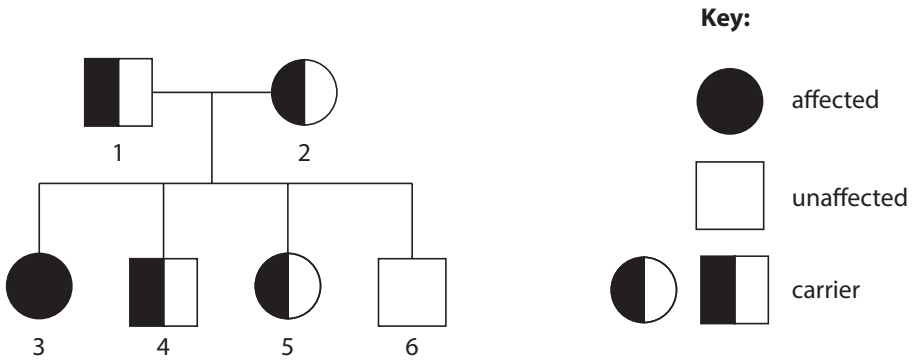
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10. Question 8 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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P 7 4 6 3 0 A 0 2 2 2 4

10. Question 8 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

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(iv) Draw a genetic diagram to show how two parents can have a 1 in 4 chance of having a child with cystic fibrosis.

Use F for the non-cystic fibrosis allele and f for the cystic fibrosis allele.

(4)

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(b) Describe how gene therapy can be used to treat cystic fibrosis.

(3)

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(Total for Question 8 = 11 marks)

10. Question 8 · part a.iv · 4 marks · Working and answer

Working and answer

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11. Question 6 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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6 (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

Region	Total number of people tested	Percentage of colour blind males	Percentage of colour blind females
A	150	10.00	1.00
B	16 000	6.60	0.40
C	5000	0.80	0.01

(i) Calculate the number of colour blind males in the test group in region C.

(3)

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number of males = ...

(ii) State why the results from region A are not considered to be reliable.

(1)

.....

P 7 4 6 2 8 A 0 1 5 2 0

11. Question 6 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

(iii) Justify the statement that colour blindness is a recessive sex-linked condition.

(5)

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11. Question 6 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

(b) A woman who is heterozygous for the red-green colour blind condition has a child. The father does not have the condition.

Use a genetic diagram to determine the possible genotypes and phenotypes of the child.

Use B to represent the allele for normal vision and b to represent the allele for the colour blind condition.

(4)

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(Total for Question 6 = 13 marks)

11. Question 6 · part b · 4 marks · Working and answer

Working and answer

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12. Question 2 · part b.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

2 (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

(1)

(ii) Which of these, found in the mid-piece, release energy?

(1)

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

(3)

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12. Question 2 · part b.i · 3 marks

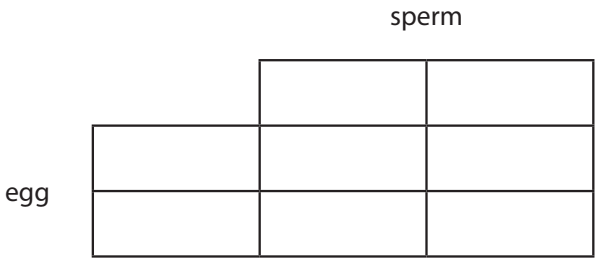
January 2023 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Complete the genetic diagram to show how the sex of a person is determined by the chromosomes in the nucleus of the egg and the nucleus of the sperm.

(3)



(ii) The sperm head has to enter the egg cell so the nuclei can fuse.

Describe how the sperm head enters the egg cell.

(2)

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(Total for Question 2 = 10 marks)

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12. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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Practice workbook · Explain sex-linked inheritance and why males are affected more

X-linked recessive conditions such as haemophilia and colour blindness, contrasted with Y-linked traits.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-SEX-LINKED-INHERITANCE

1. Question 7 · part b · 4 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b only.

7 The four blood groups that occur in humans are group A, group B, group AB and group O.

There are three blood group alleles: I^A, I^B and I^O.

I^O is recessive to both I^A and I^B.

I^A and I^B are co-dominant to each other.

(a) (i) Explain what is meant by the term **co-dominant**.

(2)

(ii) State the possible genotypes of a person with blood group B.

(2)

(b) Haemophilia is a sex-linked condition that prevents the blood clotting.

Explain why males are more likely to be affected by haemophilia than females.

(4)

(Total for Question 7 = 8 marks)

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1. Question 7 · part b · 4 marks · Working and answer

Working and answer

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2. Question 6 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

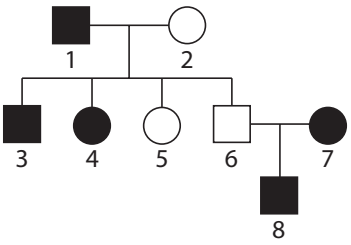
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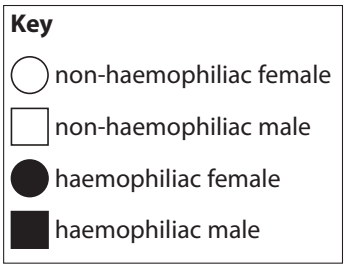
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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2. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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3. Question 6 · part b.iv · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

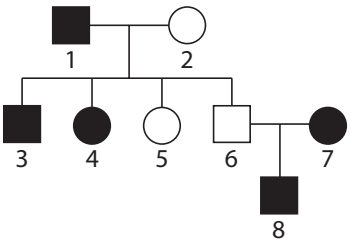
(3)

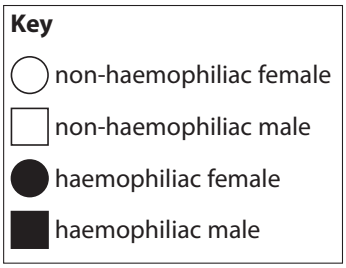
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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3. Question 6 · part b.iv · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

(ii) What is the genotype of person 5?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

(2)

probability = ...

(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

(3)

.....

.....

.....

.....

(Total for Question 6 = 12 marks)

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3. Question 6 · part b.iv · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 8 · part c · 5 marks

November 2020 Paper 2

Further practice · Attempt part c only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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4. Question 8 · part c · 5 marks

November 2020 Paper 2

Further practice · Attempt part c only.

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(ii) The man and woman have a second child.

Calculate the probability that both children will have haemophilia.

(3)

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(c) Explain why the man cannot be a carrier of haemophilia.

(5)

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(Total for Question 8 = 15 marks)

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4. Question 8 · part c · 5 marks · Working and answer

Working and answer

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5. Question 9 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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5. Question 9 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

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(b) Haemophilia is a sex-linked genetic condition that affects the blood.

(i) Describe what is meant by a sex-linked genetic condition.

(2)

.....

.....

.....

.....

(ii) State one effect that haemophilia will have on an individual.

(1)

.....

.....

.....

(Total for Question 9 = 10 marks)

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Working and answer

6. Question 5 · part b.i · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.i only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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6. Question 5 · part b.i · 2 marks

November 2024 Paper 2

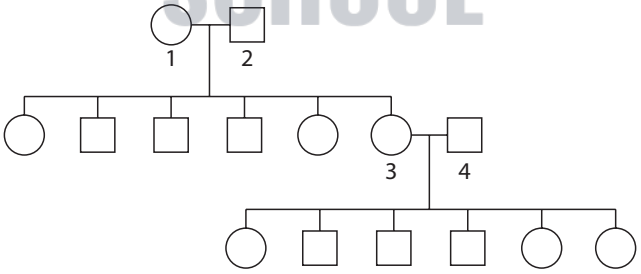
Further practice · Attempt part b.i only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

(2)

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.



Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

(2)

person 1

person 2

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6. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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7. Question 6 · part a.iii · 5 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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6 (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

Region	Total number of people tested	Percentage of colour blind males	Percentage of colour blind females
A	150	10.00	1.00
B	16 000	6.60	0.40
C	5000	0.80	0.01

(i) Calculate the number of colour blind males in the test group in region C.

(3)

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number of males = ...

(ii) State why the results from region A are not considered to be reliable.

(1)

.....

7. Question 6 · part a.iii · 5 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

(iii) Justify the statement that colour blindness is a recessive sex-linked condition.

(5)

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7. Question 6 · part a.iii · 5 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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(b) A woman who is heterozygous for the red-green colour blind condition has a child. The father does not have the condition.

Use a genetic diagram to determine the possible genotypes and phenotypes of the child.

Use B to represent the allele for normal vision and b to represent the allele for the colour blind condition.

(4)

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(Total for Question 6 = 13 marks)

7. Question 6 · part a.iii · 5 marks · Working and answer

Working and answer

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Practice workbook · Deduce genotypes from a pedigree, family or blood group

Assigning genotypes to people in family trees, including X-linked and Y-linked symbols, possible ABO genotypes for a blood group, and matching children to parents.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEDUCE-GENOTYPES-FROM-PEDIGREE

1. Question 5 · part b.ii · 1 mark

November 2023 Paper 2

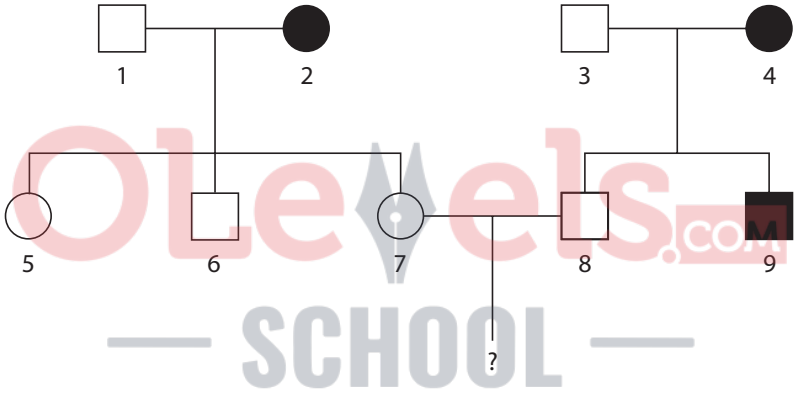
Worked-example retry · Attempt part b.ii only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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P 7 4 4 9 4 A 0 1 2 2 0

1. Question 5 · part b.ii · 1 mark

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

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(c) (i) People 7 and 8 in the pedigree produce a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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(ii) Calculate the probability that this child will be a boy with sickle cell anaemia.

(4)

probability = ...

(Total for Question 5 = 13 marks)

1. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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2. Question 9 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

(3)

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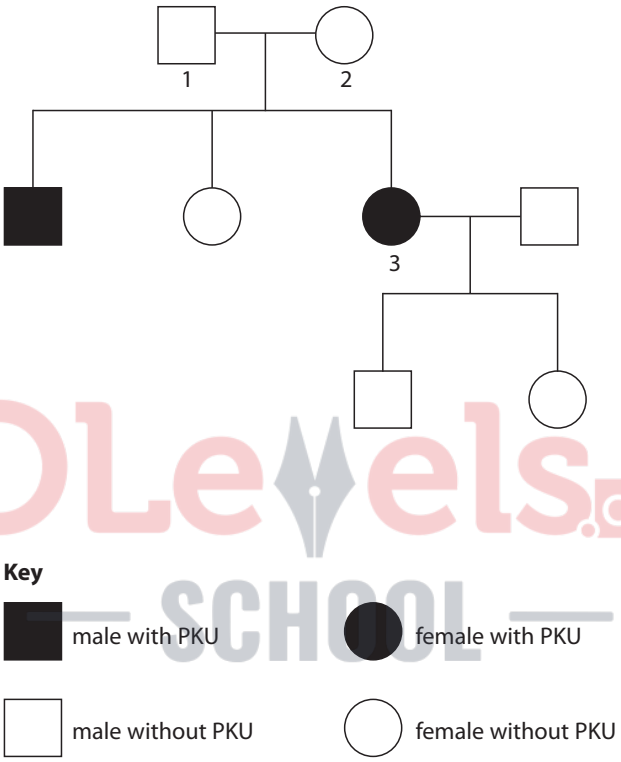
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2. Question 9 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

(b) The diagram shows how PKU is inherited in a family.



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(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

(1)

.....

(ii) Determine the probability that person 2 has one allele for PKU.

(1)

.....

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2. Question 9 · part b.i · 1 mark · Working and answer

Working and answer

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3. Question 9 · part b.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.ii only.

9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

(3)

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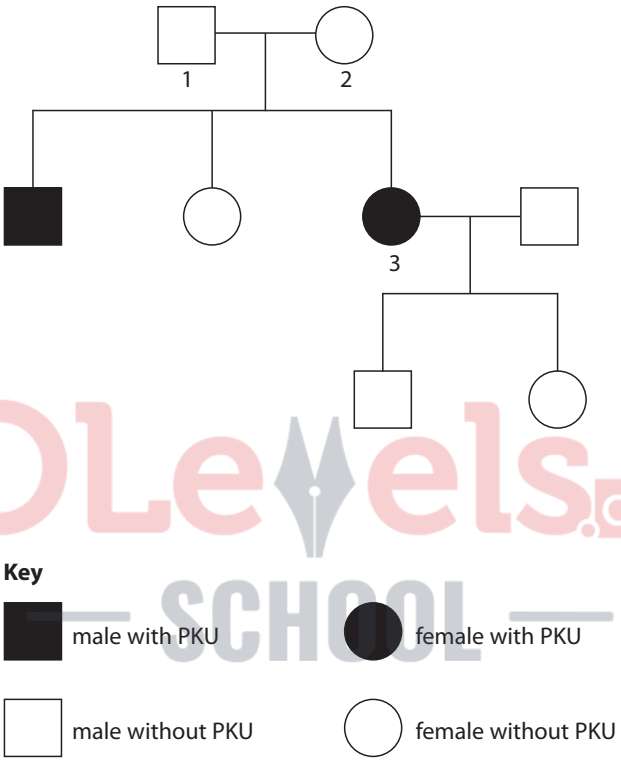
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3. Question 9 · part b.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.ii only.

(b) The diagram shows how PKU is inherited in a family.



(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

(1)

(ii) Determine the probability that person 2 has one allele for PKU.

(1)

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3. Question 9 · part b.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Deduce the genotype of person 3.

You may draw a diagram to support your answer.

(3)

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(Total for Question 9 = 14 marks)

Working and answer

4. Question 9 · part b.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part b.iii only.

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9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

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(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

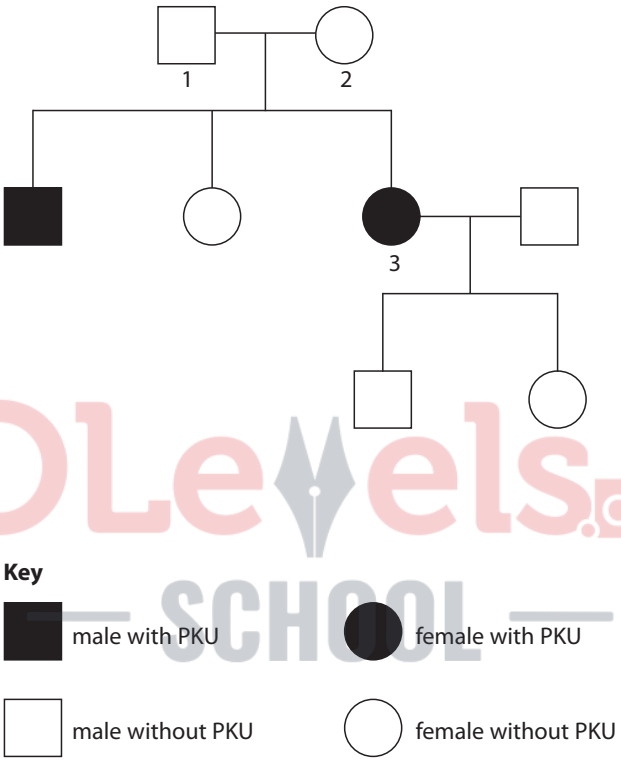
(3)

4. Question 9 · part b.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part b.iii only.

(b) The diagram shows how PKU is inherited in a family.



(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

(1)

(ii) Determine the probability that person 2 has one allele for PKU.

(1)

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4. Question 9 · part b.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part b.iii only.

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(iii) Deduce the genotype of person 3.

You may draw a diagram to support your answer.

(3)

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(Total for Question 9 = 14 marks)

4. Question 9 · part b.iii · 3 marks · Working and answer

Working and answer

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5. Question 6 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

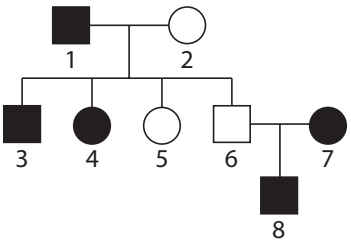
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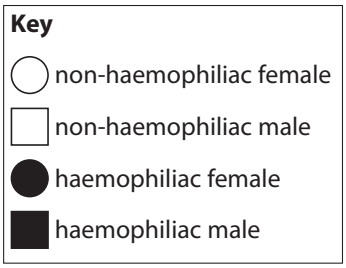
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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5. Question 6 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

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(ii) What is the genotype of person 5?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

(2)

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probability = ...

(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

(3)

.....

.....

.....

.....

(Total for Question 6 = 12 marks)

5. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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6. Question 6 · part b.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

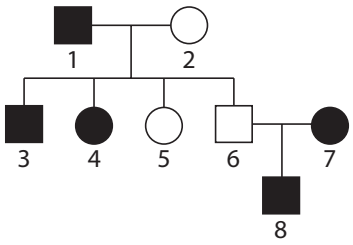
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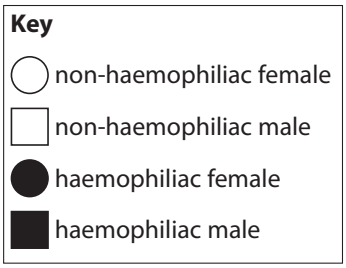
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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6. Question 6 · part b.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) What is the genotype of person 5?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

(2)

probability = ...

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(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

(3)

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(Total for Question 6 = 12 marks)

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6. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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7. Question 6 · part b.i · 5 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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6 Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

(2)

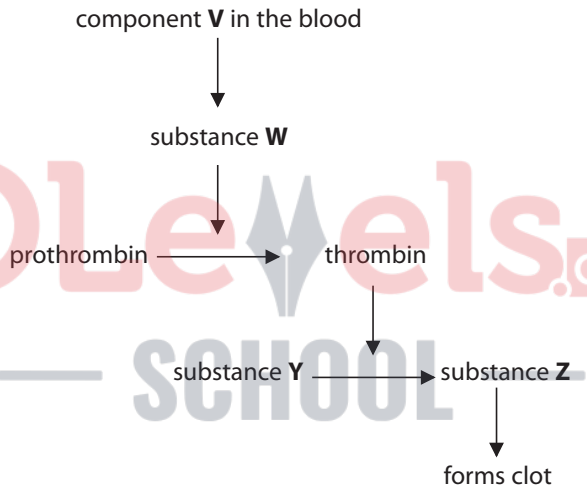
1 .....

.....

2 .....

.....

(ii) The diagram shows some of the stages in the process of blood clotting.



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Use words from the box to identify V, W, Y and Z.

thrombokinase enzyme		white blood cells		platelets
haemoglobin	fibrinogen	fibrin	antigens	antibody

(4)

V .....

W .....

Y .....

Z .....

P 7 0 9 5 0 A 0 1 7 2 4

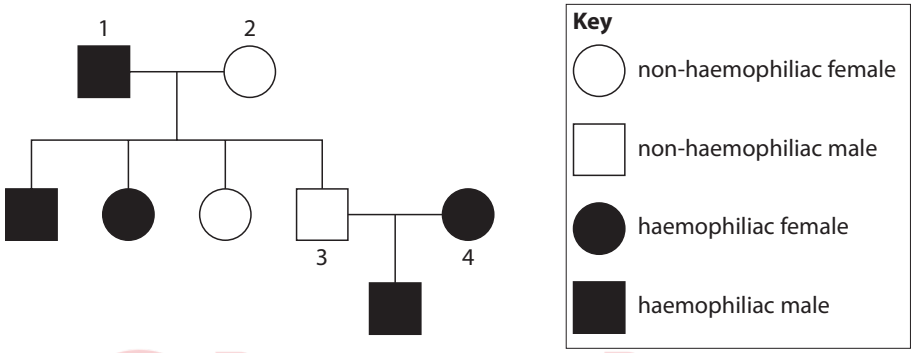
7. Question 6 · part b.i · 5 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

(b) Haemophilia is an inherited condition where the blood of a person does not clot.

The diagram shows a family pedigree. Some members of the family have the condition.



(i) Deduce the genotypes of family members 1 and 2 showing how you obtained your answer. **(5)**

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7. Question 6 · part b.i · 5 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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(ii) Male 3 and female 4 have a second child.

Deduce the probability that this child will be a boy with haemophilia showing how you obtained your answer.

(4)

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(Total for Question 6 = 15 marks)

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7. Question 6 · part b.i · 5 marks · Working and answer

Working and answer

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8. Question 7 · part c.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part c.i only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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8. Question 7 · part c.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part c.i only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

(2)

percentage = ... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

(2)

1 .....

.....

2 .....

.....

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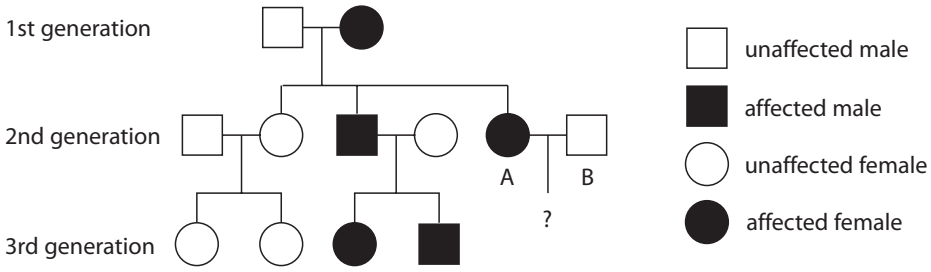
8. Question 7 · part c.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part c.i only.

(c) The diagram shows a family pedigree. Some of the family have polycystic kidney disease.

This disease is caused by a dominant allele that is not sex-linked.



(i) Determine the ratio of the females shown in the 3rd generation that are affected with polycystic kidney disease.

(1)

ratio = ...

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8. Question 7 · part c.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part c.i only.

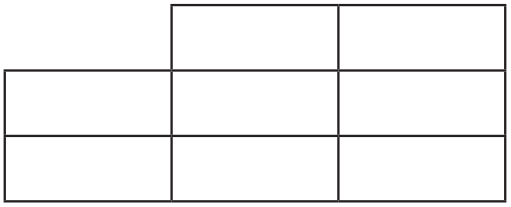
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(ii) Explain the possible genotypes and phenotypes for the offspring of individual A and individual B.

You should complete the genetic diagram as part of your answer.

(4)



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(Total for Question 7 = 14 marks)

Working and answer

9. Question 6 · part b · 4 marks

June 2023 Paper 2

Further practice · Attempt part b only.

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6 Sickle cell anaemia is an inherited condition. It is caused by a mutation of the gene that controls the production of haemoglobin.

This condition is caused by a recessive allele.

(a) (i) Explain how a mutation can cause a condition such as sickle cell anaemia.

(4)

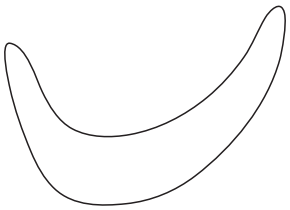
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9. Question 6 · part b · 4 marks

June 2023 Paper 2

Further practice · Attempt part b only.

(ii) In low oxygen concentrations the red blood cells of someone with sickle cell anaemia become like the cell shown in the diagram.



Explain why a person who has the sickle cell condition often feels tired.

(5)

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9. Question 6 · part b · 4 marks

June 2023 Paper 2

Further practice · Attempt part b only.

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(b) A woman, heterozygous for sickle cell anaemia, and a man of unknown genotype for sickle cell anaemia have a child. The child is born with sickle cell anaemia.

Explain the possible genotypes of the man.

Use H to represent the allele for unaffected haemoglobin and h to represent the allele for sickle haemoglobin.

(4)

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(Total for Question 6 = 13 marks)

9. Question 6 · part b · 4 marks · Working and answer

Working and answer

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10. Question 5 · part b.i · 2 marks

November 2023 Paper 2

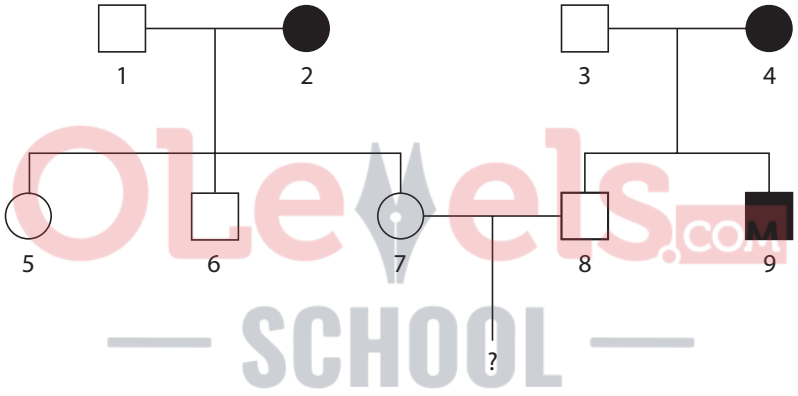
Further practice · Attempt part b.i only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



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Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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10. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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11. Question 4 · part c.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.i only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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P 7 4 4 9 5 A 0 1 2 2 8

11. Question 4 · part c.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.i only.

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(ii) Explain why diploid cells contain two alleles for each characteristic.

(4)

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

(2)

(ii) Give one symptom of cystic fibrosis.

(1)

(Total for Question 4 = 11 marks)

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11. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

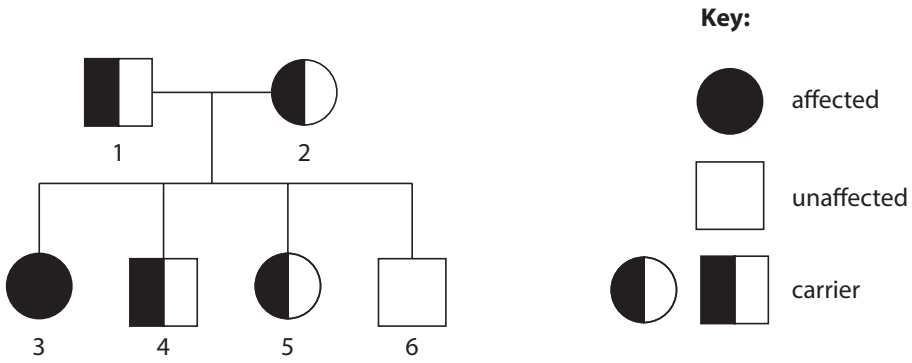
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12. Question 8 · part a.i · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.i only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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12. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

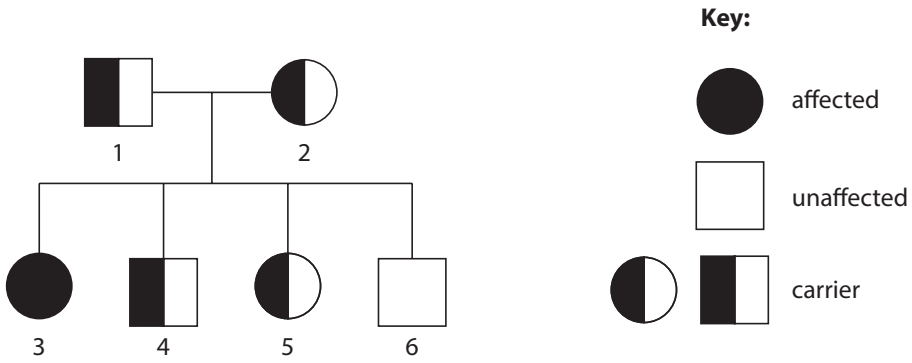
OLevels.com SCHOOL

13. Question 8 · part a.ii · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.ii only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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13. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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14. Question 7 · part a.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.i only.

7 (a) Two children, M and N, are born to two sets of parents.

Child M has blood group O and child N has blood group A.

The blood groups of the two sets of parents are given in the table.

	Blood group	
	mother	**father**
1st set of parents	AB	O
2nd set of parents	B	B

(i) Explain which child belongs to which set of parents.

(4)

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(ii) Calculate the probability that the next child produced by the 2nd set of parents will have blood group B.

Include a genetic diagram in your answer.

(3)

probability = ...

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14. Question 7 · part a.i · 4 marks · Working and answer

Working and answer

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15. Question 5 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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15. Question 5 · part b.ii · 2 marks

November 2024 Paper 2

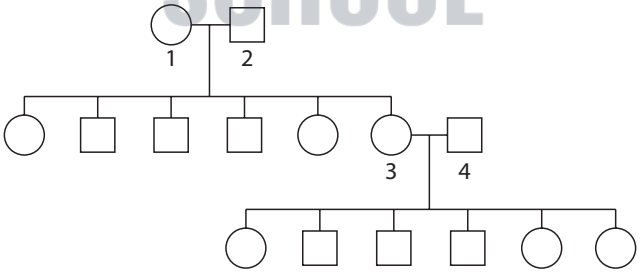
Further practice · Attempt part b.ii only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

(2)

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.



Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

(2)

person 1

person 2

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15. Question 5 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Neither person 3 nor person 4 has the condition.

Explain why the probability of their next child being a girl without hairy ears is 50%.

(3)

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(Total for Question 5 = 12 marks)

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Working and answer

Practice workbook · Build or read a pedigree to decide how a condition is inherited

Drawing a family tree from a description, and using pedigree patterns to judge dominant versus recessive and autosomal versus sex-linked inheritance.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INFER-INHERITANCE-MODE-FROM-PEDIGREE

1. Question 8 · part b.i · 4 marks

June 2019 Paper 2R

Worked-example retry · Attempt part b.i only.

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8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

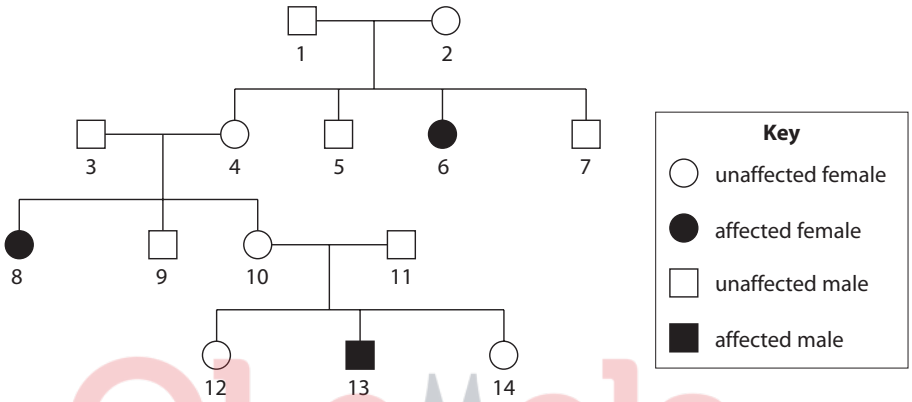
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1. Question 8 · part b.i · 4 marks

June 2019 Paper 2R

Worked-example retry · Attempt part b.i only.

(b) The diagram shows the pedigree of a family affected by alkaptonuria.



(i) Use the family pedigree to explain whether the alkaptonuria condition is sex-linked.

(4)

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1. Question 8 · part b.i · 4 marks

June 2019 Paper 2R

Worked-example retry · Attempt part b.i only.

(ii) Parents 10 and 11 have another child.

Calculate the probability that this child would be an affected male.

Explain your answer.

(3)

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(Total for Question 8 = 13 marks)

1. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

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2. Question 8 · part b.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

8 (a) Genetic conditions are often caused by recessive alleles.

(i) Which of these describes an allele?

(1)

- ☒ **A** a chromosome that controls a characteristic
- ☒ **B** an alternative form of a gene
- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

.....

.....

(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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P 6 7 0 6 2 R A 0 2 6 2 8

2. Question 8 · part b.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(ii) Discuss the conclusions the student can make about the inheritance of the condition.

(3)

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(Total for Question 8 = 8 marks)

Working and answer

3. Question 7 · part a.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

7 Huntington's chorea is a dominant genetic condition that results in the nervous system not working correctly. It is not a sex-linked condition.

Diagram 1 shows a family pedigree. Some members of the family have the condition.

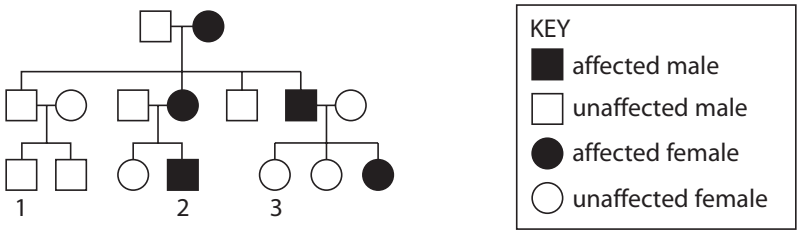


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 7 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

7 Huntington's chorea is a dominant genetic condition that results in the nervous system not working correctly. It is not a sex-linked condition.

Diagram 1 shows a family pedigree. Some members of the family have the condition.

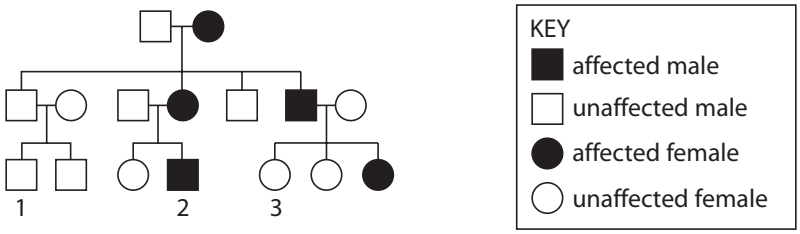


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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4. Question 7 · part a.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

(b) (i) A man has the same genotype as person 1 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

These two people are expecting a child.

Calculate the probability that the child would be a boy not showing the Huntington's condition, giving reasons for your answer.

(3)

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4. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 8 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

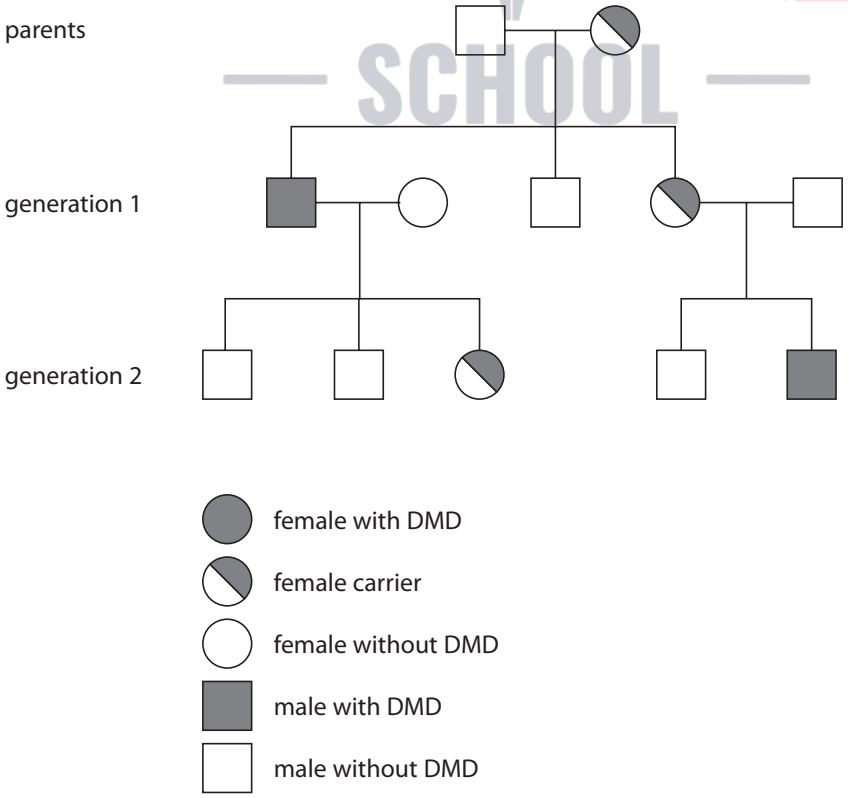
8 (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

(4)

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.



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P 7 0 9 7 4 A 0 1 6 2 0

5. Question 8 · part b.ii · 3 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

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(i) Explain what is meant by a carrier of DMD.

(2)

(ii) Explain how the diagram shows that DMD is caused by a recessive allele.

(3)

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(iii) Draw a genetic diagram to show the inheritance of DMD from the parents to generation 1.

Use X^D for the dominant allele and X^d for the recessive allele.

(2)

(Total for Question 8 = 11 marks)

5. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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6. Question 3 · part b.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.ii only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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6. Question 3 · part b.ii · 1 mark

June 2024 Paper 1

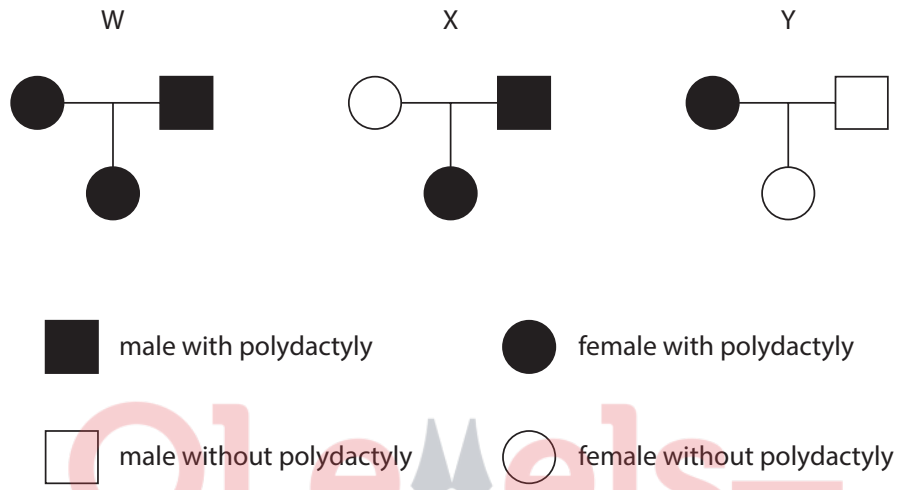
Further practice · Attempt part b.ii only.

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(ii) The family pedigree diagrams show the inheritance of polydactyly.



Which of the diagrams, W, X or Y, gives the family pedigree for parents R and S?

(1)

(c) A gene is a short section of DNA.

Describe the structure of DNA.

(4)

(Total for Question 3 = 9 marks)

6. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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7. Question 5 · part b.ii · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

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(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

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P 7 5 9 5 8 A 0 1 2 2 4

7. Question 5 · part b.ii · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

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(b) (i) Haemolytic anaemia (HA) is an inherited condition.
People with this condition have a low number of red blood cells in their blood.

Explain one symptom that people with HA will have.

(4)

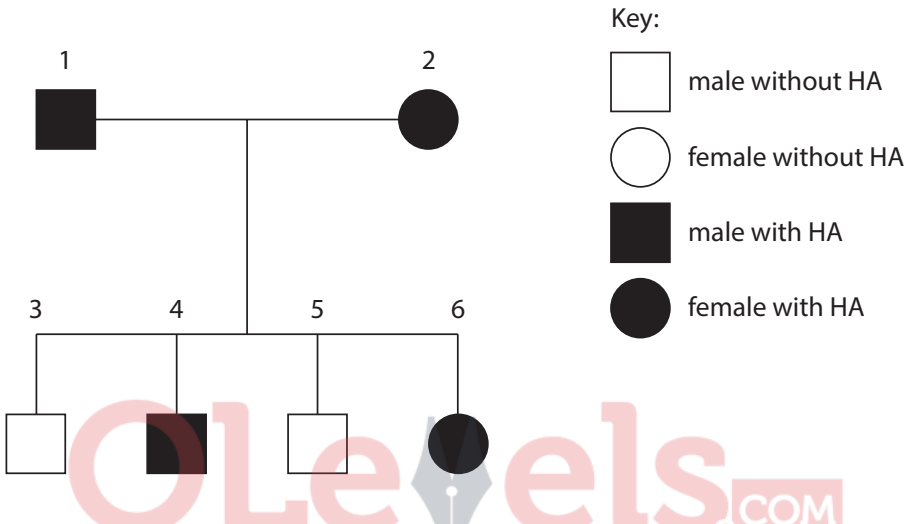
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7. Question 5 · part b.ii · 4 marks

November 2024 Paper 1

Further practice · Attempt part b.ii only.

(ii) The diagram shows the inheritance of HA in one family.



Explain how HA is inherited.

(4)

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(Total for Question 5 = 14 marks)

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7. Question 5 · part b.ii · 4 marks · Working and answer

Working and answer

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8. Question 8 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

8 (a) Genetic conditions are often caused by recessive alleles.

(i) Which of these describes an allele?

(1)

- ☒ **A** a chromosome that controls a characteristic
- ☒ **B** an alternative form of a gene
- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

.....

.....

(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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8. Question 8 · part b.i · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

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(ii) Discuss the conclusions the student can make about the inheritance of the condition.

(3)

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(Total for Question 8 = 8 marks)

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Working and answer

Practice workbook · Work out the probability of a child's genotype or phenotype from a cross

Single outcomes read from a Punnett square and combined outcomes such as an affected boy or two affected children, for autosomal, sex-linked and blood-group crosses.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PROBABILITY-FROM-A-CROSS

1. Question 7 · part a.ii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.ii only.

7 (a) Two children, M and N, are born to two sets of parents.

Child M has blood group O and child N has blood group A.

The blood groups of the two sets of parents are given in the table.

	Blood group	
	mother	**father**
1st set of parents	AB	O
2nd set of parents	B	B

(i) Explain which child belongs to which set of parents.

(4)

(ii) Calculate the probability that the next child produced by the 2nd set of parents will have blood group B.

Include a genetic diagram in your answer.

(3)

probability = ...

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1. Question 7 · part a.ii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.ii only.

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(b) Describe how to prepare a sample of blood to be observed using a light microscope.

(5)

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(Total for Question 7 = 12 marks)

1. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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2. Question 9 · part c.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part c.ii only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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2. Question 9 · part c.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part c.ii only.

(c) Haemophilia is a sex-linked blood disorder that reduces the ability of the blood to clot.

These are the genotypes of four offspring, P, Q, R and S.

P	Q	R	S
X^HX^h	X^hY	X^HX^H	X^HY

(i) Draw a genetic diagram to show how these offspring are produced from one set of parents.

(2)

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(ii) These parents are expecting another baby.

Determine the probability that this baby will have haemophilia.

(1)

probability = ...

(Total for Question 9 = 9 marks)

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2. Question 9 · part c.ii · 1 mark · Working and answer

Working and answer

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3. Question 6 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

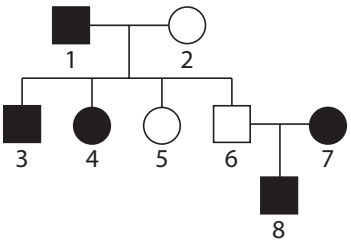
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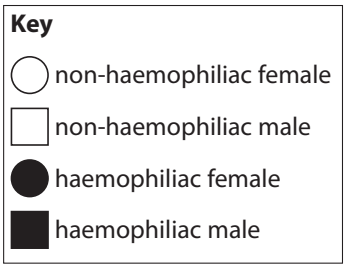
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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3. Question 6 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

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(ii) What is the genotype of person 5?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

(2)

probability = ...

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(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

(3)

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(Total for Question 6 = 12 marks)

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3. Question 6 · part b.iii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part a.iv · 1 mark

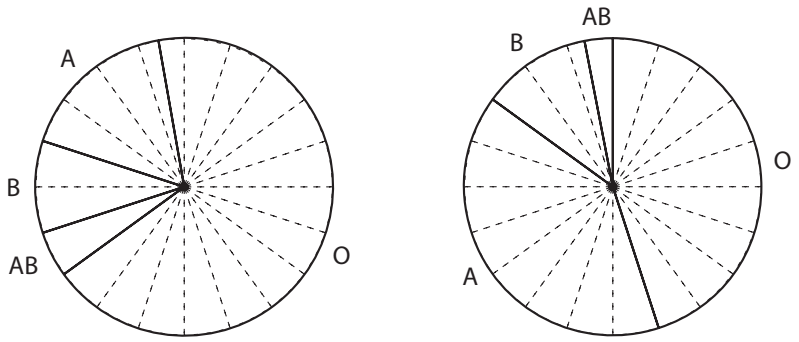
June 2022 Paper 1

Further practice · Attempt part a.iv only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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P 7 0 9 7 3 A 0 1 2 2 4

4. Question 4 · part a.iv · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.iv only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...

4. Question 4 · part a.iv · 1 mark

June 2022 Paper 1

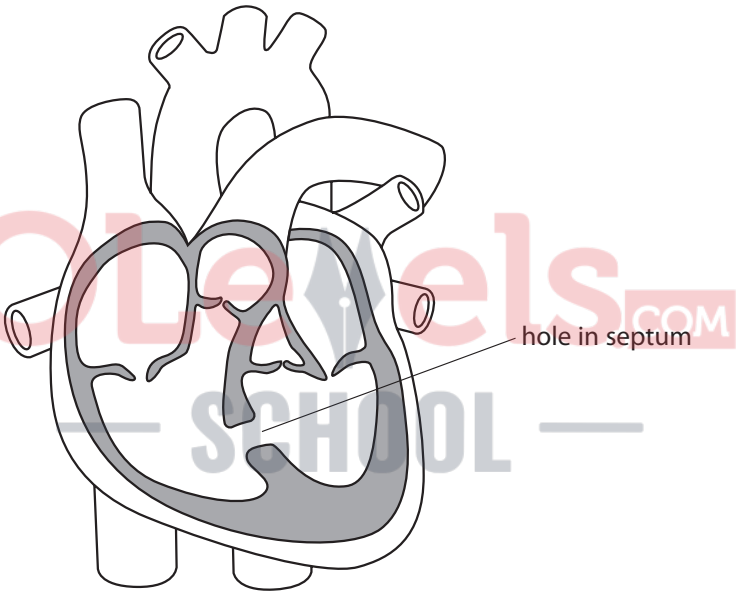
Further practice · Attempt part a.iv only.

(b) The heart pumps the blood around the body.

The diagram shows a human heart.

This heart has a hole in the septum.

This means that blood from the left side of the heart mixes with blood from the right side.



Explain why a person with a hole in their heart breathes at a greater rate than a person with a healthy heart.

(4)

(Total for Question 4 = 13 marks)

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4. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

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5. Question 6 · part b.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

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6 Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

(2)

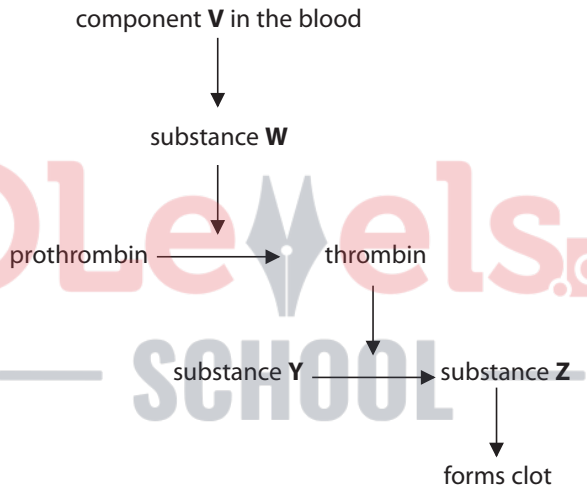
1 .....

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2 .....

.....

(ii) The diagram shows some of the stages in the process of blood clotting.



Use words from the box to identify V, W, Y and Z.

thrombokinase enzyme		white blood cells		platelets
haemoglobin	fibrinogen	fibrin	antigens	antibody

(4)

V .....

W .....

Y .....

Z .....

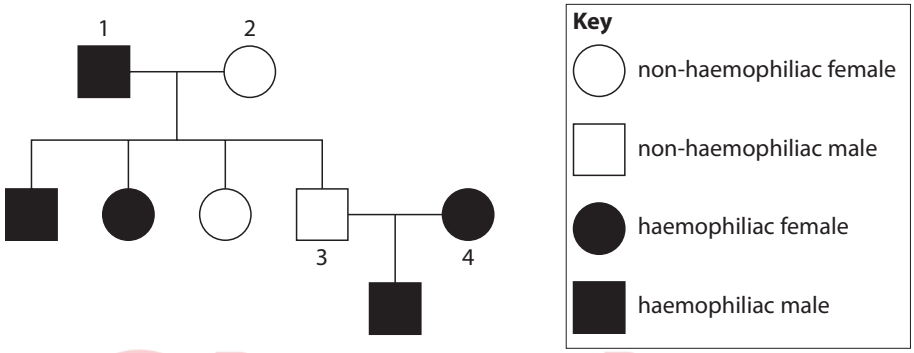
5. Question 6 · part b.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

(b) Haemophilia is an inherited condition where the blood of a person does not clot.

The diagram shows a family pedigree. Some members of the family have the condition.



(i) Deduce the genotypes of family members 1 and 2 showing how you obtained your answer.

(5)

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5. Question 6 · part b.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Male 3 and female 4 have a second child.

Deduce the probability that this child will be a boy with haemophilia showing how you obtained your answer.

(4)

(Total for Question 6 = 15 marks)

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5. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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6. Question 8 · part b.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

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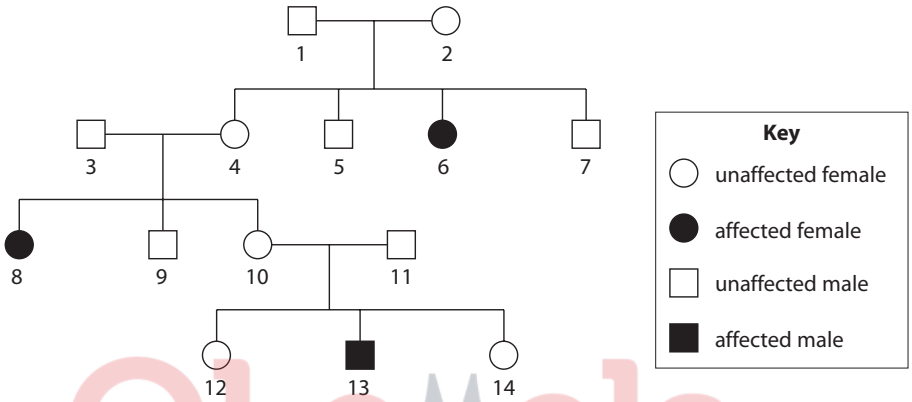
OLevels.COM SCHOOL

6. Question 8 · part b.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

(b) The diagram shows the pedigree of a family affected by alkaptonuria.



(i) Use the family pedigree to explain whether the alkaptonuria condition is sex-linked.

(4)

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6. Question 8 · part b.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Parents 10 and 11 have another child.

Calculate the probability that this child would be an affected male.

Explain your answer.

(3)

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(Total for Question 8 = 13 marks)

6. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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7. Question 8 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

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(ii) The man and woman have a second child.

Calculate the probability that both children will have haemophilia.

(3)

(c) Explain why the man cannot be a carrier of haemophilia.

(5)

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(Total for Question 8 = 15 marks)

7. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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8. Question 8 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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8. Question 8 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Calculate the probability that these parents will produce a boy with the condition.

(3)

probability = ...

(c) White forelock is a result of a mutation in a molecule of DNA.

Describe how this mutation causes the white forelock phenotype.

(4)

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(Total for Question 8 = 14 marks)

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8. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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9. Question 7 · part b.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

7 Huntington's chorea is a dominant genetic condition that results in the nervous system not working correctly. It is not a sex-linked condition.

Diagram 1 shows a family pedigree. Some members of the family have the condition.

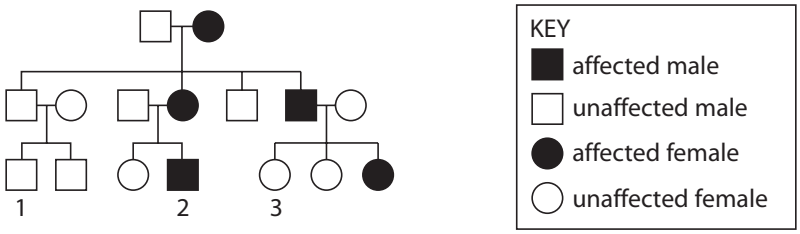


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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9. Question 7 · part b.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

(b) (i) A man has the same genotype as person 1 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

These two people are expecting a child.

Calculate the probability that the child would be a boy not showing the Huntington's condition, giving reasons for your answer.

(3)

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9. Question 7 · part b.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

(ii) A man has the same genotype as person 2 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

Explain the possible phenotypes and genotypes, with respect to the Huntington's condition, of any offspring they produce.

(6)

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(Total for Question 7 = 13 marks)

9. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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10. Question 5 · part c.ii · 4 marks

November 2023 Paper 2

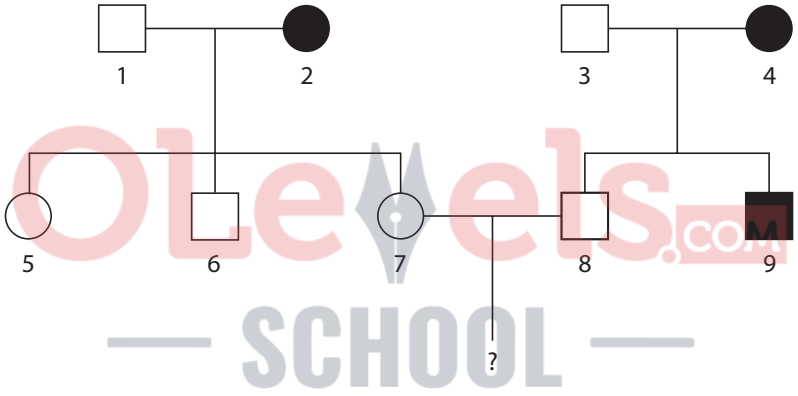
Further practice · Attempt part c.ii only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



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Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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10. Question 5 · part c.ii · 4 marks

November 2023 Paper 2

Further practice · Attempt part c.ii only.

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(c) (i) People 7 and 8 in the pedigree produce a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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(ii) Calculate the probability that this child will be a boy with sickle cell anaemia.

(4)

probability = ...

(Total for Question 5 = 13 marks)

10. Question 5 · part c.ii · 4 marks · Working and answer

Working and answer

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11. Question 5 · part b.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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11. Question 5 · part b.iii · 3 marks

November 2024 Paper 2

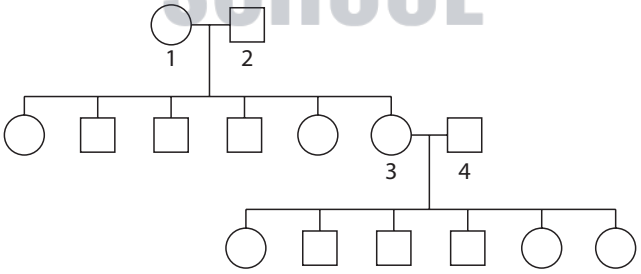
Further practice · Attempt part b.iii only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

(2)

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.



Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

(2)

person 1

person 2

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11. Question 5 · part b.iii · 3 marks

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Further practice · Attempt part b.iii only.

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(iii) Neither person 3 nor person 4 has the condition.

Explain why the probability of their next child being a girl without hairy ears is 50%.

(3)

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(Total for Question 5 = 12 marks)

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Working and answer